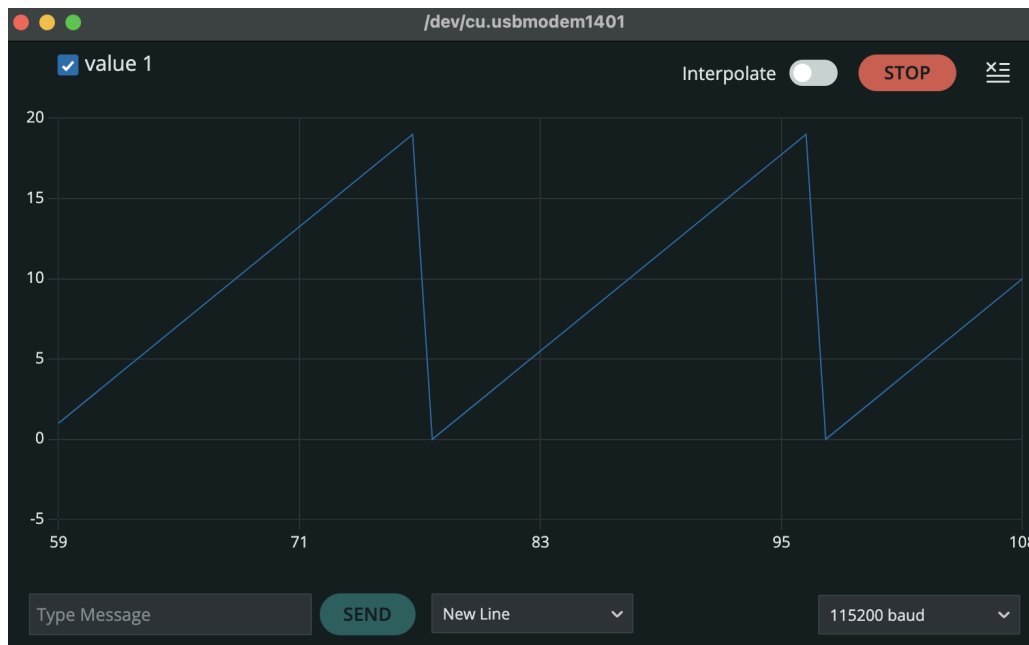


Task 4:

Serial Monitor:

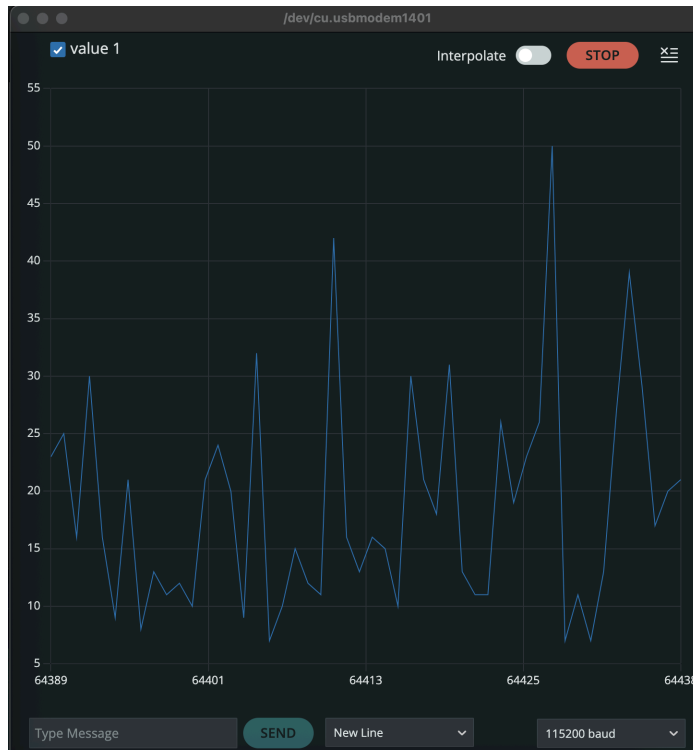


Serial Plotter:

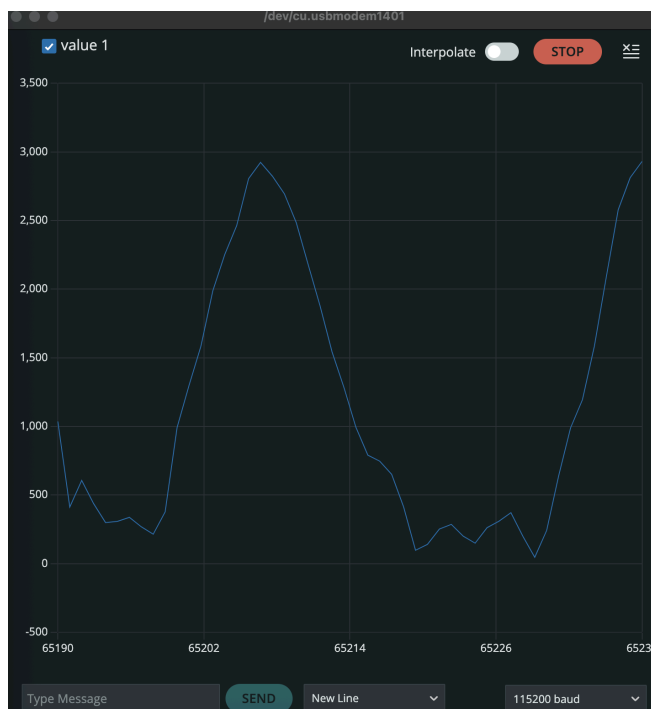


Task 5:

Serial Plotter Quiet:



Serial Plotter Loud:

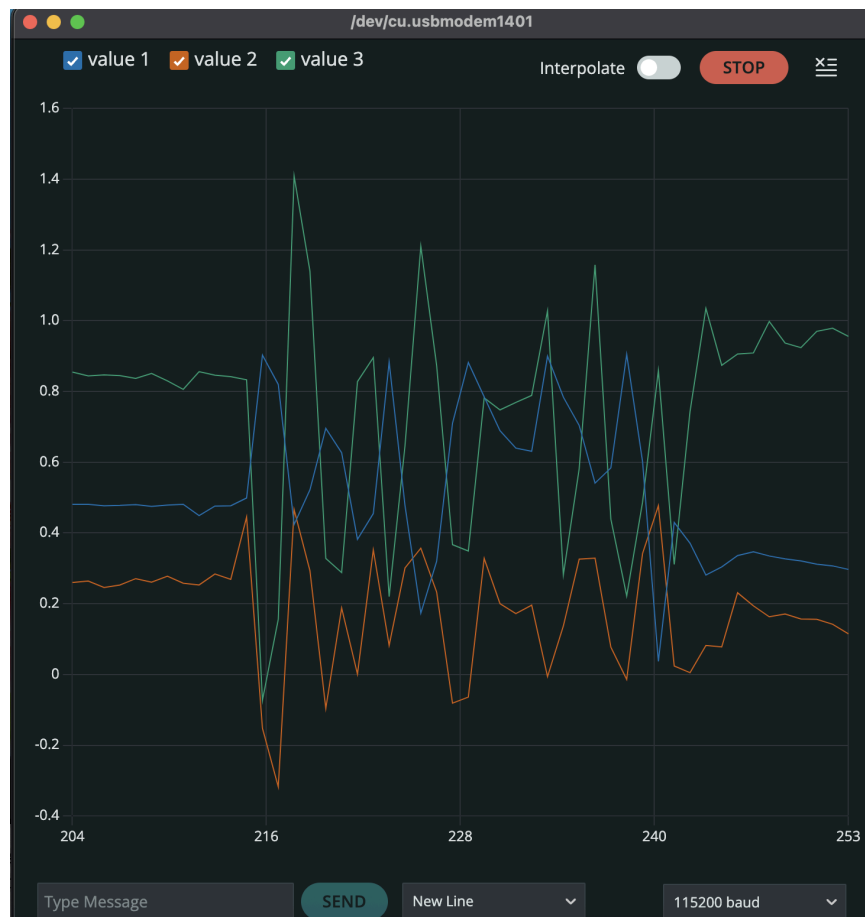


The loud plot has significantly higher amplitudes (~ 3,000) compared to the quiet plot which has amplitudes of ~40-50.

Task 6A Serial Monitor:



Task 6A Serial Plotter:



The movement that produced the most change was drawing a figure “8” with the arduino. I think was because there was inclusion of up, down, left, and right movements.

Task 6B Serial Monitor:



The screenshot shows the Arduino IDE Serial Monitor window. The title bar includes 'Output', 'Serial Monitor', and a close button. The interface features a text input field with the placeholder 'Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev...)', a 'New Line' button, and a baud rate dropdown set to '115200 baud'. The main area displays a list of 10 coordinate pairs, each on a new line. A status bar at the bottom indicates 'Ln 1, Col 1' and the device 'Arduino Nano 33 BLE on /dev/cu.usbmodem1401'.

```
0.314,-1.679,0.293
0.510,3.721,1.448
0.397,-2.311,-0.181
0.672,3.652,1.619
0.512,-0.650,-0.037
0.483,1.094,0.652
0.250,1.137,0.195
0.432,0.249,0.096
0.311,0.994,0.592
0.293,0.766,0.091
```

Task 6B Serial Plotter:



The movement that produced the most change was vigorously shaking the board front, back, and then up, and down.

Task 6C Serial Monitor:

```
Output  Serial Monitor X
Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/ New Line 115
5.000,-15.000,17.000
5.000,-15.000,14.000
14.000,-14.000,18.000
0.000,-4.000,13.000
-12.000,0.000,7.000
-16.000,-1.000,9.000
-12.000,-2.000,9.000
-13.000,0.000,11.000
-15.000,-2.000,9.000
-16.000,-1.000,9.000
Ln 1, Col 1  Arduino Nano 33 BLE on /dev/cu.usbmodem
```

Task 6C Serial Monitor:



Putting the arduino nano on top of my laptop produced the greatest change with the serial monitor values and the serial plot signal graphs.

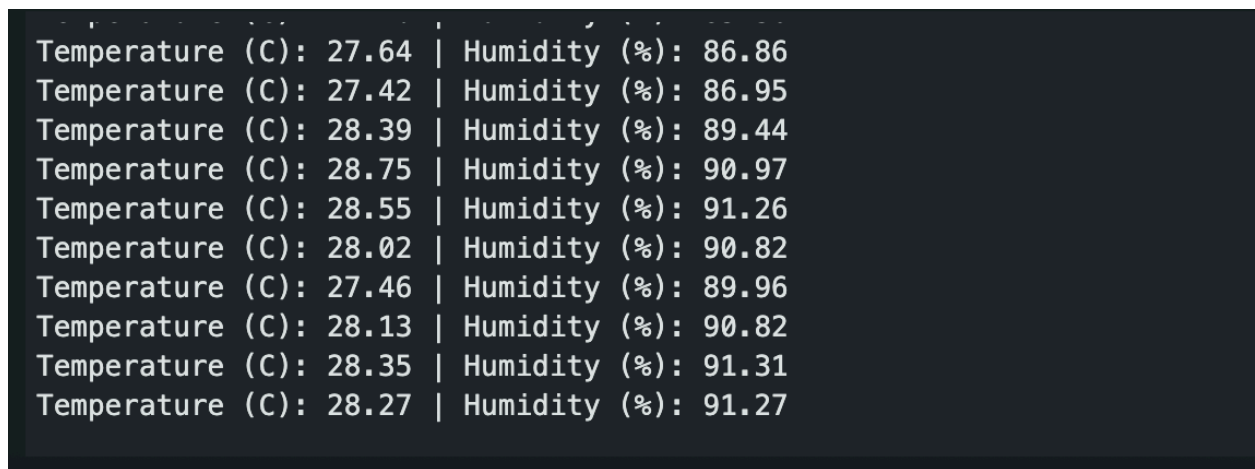
Task 7 Normal Temperature Serial Monitor:



The screenshot shows the Arduino IDE Serial Monitor window. The title bar reads "Output Serial Monitor X". The input field contains the text "Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev...". The dropdown menu is set to "New Line" and the baud rate is "115200 baud". The output text displays a series of temperature and humidity readings separated by a vertical bar.

```
Temperature (C): 25.10 | Humidity (%): 46.55
Temperature (C): 25.09 | Humidity (%): 46.49
Temperature (C): 25.07 | Humidity (%): 45.30
Temperature (C): 25.05 | Humidity (%): 44.86
Temperature (C): 25.03 | Humidity (%): 44.57
Temperature (C): 25.05 | Humidity (%): 44.05
Temperature (C): 25.05 | Humidity (%): 44.00
Temperature (C): 25.00 | Humidity (%): 43.45
Temperature (C): 25.02 | Humidity (%): 43.40
Temperature (C): 25.03 | Humidity (%): 43.72
```

Task 7 Increased temperature and humidity serial monitor:



The screenshot shows the Arduino IDE Serial Monitor window displaying a series of temperature and humidity readings. The values are significantly higher than those in the previous screenshot, indicating an increase in both temperature and humidity.

```
Temperature (C): 27.64 | Humidity (%): 86.86
Temperature (C): 27.42 | Humidity (%): 86.95
Temperature (C): 28.39 | Humidity (%): 89.44
Temperature (C): 28.75 | Humidity (%): 90.97
Temperature (C): 28.55 | Humidity (%): 91.26
Temperature (C): 28.02 | Humidity (%): 90.82
Temperature (C): 27.46 | Humidity (%): 89.96
Temperature (C): 28.13 | Humidity (%): 90.82
Temperature (C): 28.35 | Humidity (%): 91.31
Temperature (C): 28.27 | Humidity (%): 91.27
```

Lightly breathing on the arduino boosted the temperature and humidity values.

Task 8:

Pressure (kPa): 100.696	Temperature (C): 22.57
Pressure (kPa): 100.692	Temperature (C): 22.57
Pressure (kPa): 100.695	Temperature (C): 22.57
Pressure (kPa): 100.694	Temperature (C): 22.57
Pressure (kPa): 100.693	Temperature (C): 22.57
Pressure (kPa): 100.695	Temperature (C): 22.57
Pressure (kPa): 100.693	Temperature (C): 22.57
Pressure (kPa): 100.696	Temperature (C): 22.57
Pressure (kPa): 100.695	Temperature (C): 22.56
Pressure (kPa): 100.697	Temperature (C): 22.56
Pressure (kPa): 100.697	Temperature (C): 22.55

Task 9A:

Proximity: 240
Proximity: 241
Proximity: 240
Proximity: 238
Proximity: 0
Proximity: 0
Proximity: 0
Proximity: 0

Task 9B:

UP
DOWN
LEFT
RIGHT
LEFT
RIGHT
UP
DOWN
LEFT

Task 9C:

Baseline room lighting:

63,65,74,188
63,65,74,187
62,71,71,178
58,69,69,168
58,70,70,169
58,70,70,168

Colored Room lighting:

- Red:

295,309,326,863
297,312,329,872
294,309,326,862
296,311,327,864
298,313,330,872
302,317,334,883
298,313,330,872

- Blue:

```
5,30,79,109  
5,30,79,110  
5,29,76,106  
5,29,78,106  
5,29,78,108  
5,30,78,108
```

The change that was the most drastic was when I put my phone displaying a specific color near the sensor.

Task 10:

```
raw,mic=12,clear=38,motion=1.4437,prox=56  
flags,sound=0,dark=1,moving=1,near=1  
state,QUIET_BRIGHT_STEADY_FAR
```

```
raw,mic=442,clear=176,motion=1.6461,prox=242  
flags,sound=1,dark=0,moving=1,near=1  
state,NOISY_BRIGHT_MOVING_NEAR
```

```
raw,mic=12,clear=20,motion=0.1610,prox=32  
flags,sound=0,dark=1,moving=0,near=1  
state,QUIET_DARK_STEADY_NEAR
```

```
raw,mic=27,clear=116,motion=0.2015,prox=241  
flags,sound=1,dark=0,moving=0,near=0  
state,NOISY_BRIGHT_STEADY_FAR
```

I selected my thresholds by first setting random values and watching the serial monitor fluctuate and then fine tuning the values. For audio, I first tested the quiet room microphone settings and then clapped to observe how the value would change. For light, I put the board in my room, and then turned the lights off. For proximity, I put nothing near the sensor, then would put my hand or object close to the board, and then logged the values.

Task 11:

```
raw, rh=39.56, temp=23.98, mag=3.74, r=49, g=53, b=64, clear=163  
flags, humid_jump=0, temp_rise=0, mag_shift=0, light_or_color_change=0  
event, BASELINE_NORMAL
```

```
raw, rh=90.89, temp=26.58, mag=58.99, r=49, g=53, b=64, clear=163  
flags, humid_jump=1, temp_rise=1, mag_shift=1, light_or_color_change=0  
event, MAGNETIC_DISTURBANCE_EVENT
```

```
raw, rh=85.85, temp=25.34, mag=35.69, r=49, g=53, b=64, clear=163  
flags, humid_jump=1, temp_rise=0, mag_shift=0, light_or_color_change=0  
event, BREATH_OR_WARM_AIR_EVENT
```

I selected the thresholds first by using the serial monitor to watch the normal baseline values. Then I chose thresholds that are slightly above normal so I would see the changes. The cooldown logic prevents the same event from printing too fast.